Reader: I heard that higher mathematics uses the operations of differentiation and integration.

Author: These operations, as we shall see, are in essence nothing but the variations of the limit transition.

Now, let us get down to the concept of the limit of sequence. Do you know what it is?

Reader: I learned the definition of the limit of sequence. However, I doubt that I can reproduce it from memory.

Author: But you seem to "feel" this notion somehow? Probably, you can indicate which of the sequences discussed above have limits and what the value of the limit is in each case.

Son: Sequence 5 has a limit of 1. Sequence 7 has a limit of 0. Sequence 9 has a limit of 0. Sequence 13 has a limit of π.

Author: That's right. The remaining sequences have no limits.

Reader: By the way, sequence (9) is not monotonic.....

Author: Apparently, you have just remembered the end of our previous dialogue where it was stated that if a sequence is both monotonic and bounded, it has a limit.

Reader: That's correct. But isn't this a contradiction?

Author: Where do you find the contradiction? Do you think that from the statement "If a sequence is both monotonic and bounded, it has a limit" one should necessarily draw a reverse statement like "If a sequence has a limit, it must be monotonic and bounded"?

• Later we shall see that the necessary condition for a limit is only the boundedness of a sequence. The monotonicity is not mandatory at all; consider, for example, sequence (9).

V. Imp point

• Let us get back to the concept of the limit of sequence. Since you have correctly indicated the sequences that have limits, you obviously have some understanding of this concept. Could you formulate it?